

Asset Management System

Requirements | March 2026

Overview

Goal	Log assets, assign to employees, auto-generate QR on assignment. Employees see only their own currently assigned assets.
Users	Admin (IT / HR) · Employee
Constraint	Phase 1 — 100% free & open-source stack. No paid services.

Why Are We Building ?

- Assets go missing with zero accountability
- New joiner joins → IT has no idea what's available
- Employee leaves → no checklist of what to collect back
- Audits are painful — manual spreadsheet cross-checks

Stage 1 — What are We Building ?

Only these screens. Nothing more until Stage 1 is live and working.

Login Screen

- Email + password fields
- On success: Admin → Dashboard · Employee → My Assets
- Wrong credentials → inline error message
- Logout button on every page

Admin — Asset List

- Table: asset_id, asset_type, brand, model, serial_number, status
- Search by name or serial number
- Filter by status: All / Available / Assigned
- Add New Asset button

Admin — Add / Edit Asset

- Fields: asset_type, brand, model, serial_number, configuration, status
- asset_id auto-generated on save
- Edit pre-fills the same form

Admin — Assign Asset

- Only available on assets with status = Available
- Select employee from dropdown
- Set assigned date
- On confirm: status → Assigned · QR generated · PNG download

Admin — Return Asset

- Only available on assets with status = Assigned
- Confirm return → status → Available · QR invalidated

Admin — Asset as Lost

- Only available on assets with status = Assigned
- Admin logs asset as Lost → status → Lost · QR invalidated
- Admin adds a note/reason for audit trail
- Asset flagged in audit log · cannot be reassigned

Admin — Asset as Damaged

- Only available on assets with status = Assigned
- Admin logs asset as Damaged → Admin selects sub-status:
 - → Under Repair · QR deactivated · assignable once repaired
 - → Disposed · QR invalidated · asset permanently archived
- Admin adds a note/reason for audit trail
- Requires admin confirmation prompt before committing

Admin — Dashboard

- 4 KPI cards: Total · Assigned · Available · Under Repair
- Last 10 assignments table: asset, employee, date

Admin — Audit Log

- Table: timestamp, action, performed by, asset ID
- Logs: asset added, edited, assigned, returned
- Read-only. No filters in Stage 1

Employee — My Assets

- Shows only currently assigned assets for the logged-in employee
- No other employee's data is accessible at any point
- Columns: asset name, type, serial no., assigned date
- Total count at top
- Click any row → Asset Detail

Employee — Asset Detail

- All asset fields for that specific record
- QR code image displayed
- Download QR as PNG

Employee — QR Scan

- Scan via device camera
- Own asset → opens Asset Detail
- Not own asset → shows 'Contact IT' message only

Android App — Admin & Employee

- Built with React Native + TypeScript · shared codebase with web
- Consumes the same API as web · no separate backend
- Android only
- Employee screens: Login / Logout, My Assets, Asset Detail, QR Scan
- Admin screens: Login / Logout, QR Scan → Asset Detail lookup only
- Admin full management (add, assign, audit) stays on web only

Telemetry

- **Loguru:** JSON structured logs on every API request · includes user ID, endpoint, method, status code, timestamp · lightweight Python-native logger
- **GlitchTip:** captures all server + client errors · stack traces attached · environment tagged (dev/prod) · free open-source Sentry alternative
- **Uptime Kuma:** monitors service uptime · GET /health check · alerts on downtime · free self-hosted dashboard

Activity Logs

- Log every user action — sign-in, sign-up, sign-out, asset assign, return, lost, damaged
- Each log entry captures → user ID · action type · timestamp · IP address · success/failure status
- Track per-user login count · total success attempts · total failure attempts
- Failure logs and success logs stored in **separate log tables**

Auth-Specific Logs

- Sign-in success → logged with timestamp + device info
- Sign-in failure → logged with reason (wrong password, account not found, locked) · stored separately
- Sign-up success/failure → logged independently
- Consecutive failure attempts → flagged for review

Log Retention & Purge Mechanism

- General activity logs (success, info) → **auto-deleted after 1 month**
- Failure & error logs → **auto-deleted after 6 months**
- Purge job runs as a **scheduled cron task** (nightly at 00:00)
- Deletion is soft-delete first → hard-delete after 7-day grace period

GET /health

- Returns → DB status · uptime · Uptime Kuma ping status · last purge run timestamp

Auth & Access Control (integrate services that we build independently)

Auth	Handled by the centralised Auth Service with RBAC. AMS consumes it — does not implement it.
Roles	Admin · Employee — enforced at API level on every request.
Employee rule	Can only access their own currently assigned assets. No other employee's data visible at any point.

Role Access

Action	Admin	Employee
Add / Edit Assets	✓	✗
View all assets	✓	✗
Assign / Return asset	✓	✗
Generate QR code	✓ auto	✗
View own currently assigned assets only	✓	✓
View another employee's assets	✓	✗
Scan QR code	✓	✓
Audit log	✓	✗

Stage 2 — After Stage 1 Is Live

Built only after Stage 1 is shipped, stable, and real users are using it.

Auth & Users

- SSO — Google / Microsoft login

Assets

- Asset photos
- Bulk import via Excel
- Warranty expiry email alerts
- Depreciation tracking

Reports

- CSV / PDF export
- Asset utilisation charts on dashboard
- Per-employee full history report

Platform

- Email QR to employee on assignment
- Advanced telemetry — OpenTelemetry + Grafana